

Let x be an arbitrary element of $B \cup A$, i.e. $x \in B \cup A$.
 By Defn. 1, $x \in B \vee x \in A$. By 1st Commutative Law of Propositions, $x \in A \vee x \in B$.
 By Defn. 1, $x \in A \cup B$. By Defn. 3 from Section 2.1, $B \cup A \subseteq A \cup B$.

$$\therefore A \cup B = B \cup A \quad \square$$

b) Let x be an arbitrary element of $A \cap B$, i.e. $x \in A \cap B$.
 By Defn. 2, $x \in A \wedge x \in B$. By 2nd Commutative Law of Propositions, $x \in B \wedge x \in A$.
 By Defn. 2, $x \in B \cap A$. By Defn. 3 from Section 2.1, $A \cap B \subseteq B \cap A$.

Let x be an arbitrary element of $B \cap A$, i.e. $x \in B \cap A$.
 By Defn. 2, $x \in B \wedge x \in A$. By 2nd Commutative Law of Propositions, $x \in A \wedge x \in B$.
 By Defn. 2, $x \in A \cap B$. By Defn. 3 from Section 2.1, $B \cap A \subseteq A \cap B$.

$$\therefore A \cap B = B \cap A \quad \square$$

12) Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.

Solution:

Let x be an arbitrary element of $A \cup (A \cap B)$. By Defn. of OR, at least one of $x \in A$ or $x \in A \cap B$ must be True. If $x \in A$, we are done.
 Let $x \in A \cap B$. By Defn. 2, $x \in A \wedge x \in B$. By Defn. of AND, $x \in A$ and $x \in B$.
 By Defn. 3 from Section 2.1, $A \cup (A \cap B) \subseteq A$.

Let x be an arbitrary element of A , i.e. $x \in A$.
 By Defn. of OR, $x \in A \vee x \in A \cap B$. By Defn. 1, $x \in A \cup (A \cap B)$.
 By Defn. 3 from Section 2.1, $A \subseteq A \cup (A \cap B)$.

$$\therefore A \cup (A \cap B) = A \quad \square$$

13) Prove the second absorption law from Table 1 by showing that if A and B are sets, then $A \cap (A \cup B) = A$.

Solution:

Let x be an arbitrary element of $A \cap (A \cup B)$, i.e. $x \in A \cap (A \cup B)$.
 By Defn. 2, $x \in A \wedge x \in A \cup B$. By Defn. of AND, $x \in A$ and $x \in A \cup B$. By defn. 3 of Section 2.1, $A \cap (A \cup B) \subseteq A$.

$$A \cap (A \cup B) \subseteq A$$